

MANAS, Kyrgyzstan

WMO: 382200

Lat: 43.067N

Lon: 74.483E

Elev: 2090

StdP: 13.62

Time Zone: 6.00 (E06)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	1.2	-8.0	3.9	-3.5	-2.6	5.2	2.2	16.8	33.1	14.5	33.4	2.7	180	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.0	98.7	64.7	95.9	63.6	93.3	63.1	68.9	88.5	66.7	87.3	65.2	85.7	8.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.4	91.2	77.4	60.6	85.3	74.3	58.8	79.9	71.8	34.5	88.0	32.5	86.8	31.2	85.2	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.2	16.2	13.7		-10.4	103.5	6.9	2.4	-15.4	105.2	-19.4	106.6	-23.3	107.9	-28.3	109.6	
				-10.9	69.9	6.8	3.4	-15.8	72.3	-19.8	74.3	-23.6	76.2	-28.5	78.6	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	24.0	28.3	44.3	56.4	65.5	75.1	79.3	76.4	66.5	53.3	39.8	27.8	(6)	
(7)		DBStd	21.04	10.57	11.68	10.15	8.29	6.39	5.48	5.03	5.73	7.11	8.59	10.65	10.77	(7)	
(8)		HDD50	2765	806	610	227	35	0	0	0	1	65	331	690		(8)	
(9)		HDD65	5645	1271	1029	641	274	73	4	0	3	66	371	758	1155		(9)
(10)		CDD50	3929	0	1	52	226	482	752	909	819	495	167	25	1		(10)
(11)		CDD65	1331	0	0	0	15	90	305	444	357	110	9	1	0		(11)
(12)		CDH74	16754	0	0	12	290	1121	3575	5666	4463	1448	163	16	0		(12)
(13)		CDH80	8285	0	0	0	75	367	1753	3143	2342	565	38	2	0		(13)
(14)	Wind	WSAvg	6.2	4.7	5.1	6.5	7.0	7.4	7.8	7.6	7.2	6.4	5.1	4.8	4.9	(14)	
(15)	Precipitation	PrecAvg	17.3	1.2	1.3	1.7	2.5	2.2	1.3	0.8	0.5	0.6	1.7	1.8	1.4	(15)	
(16)		PrecMax	25.9	2.7	2.2	3.3	5.7	4.5	3.7	2.6	1.5	2.2	4.4	4.6	2.9	(16)	
(17)		PrecMin	11.3	0.3	0.4	0.9	0.5	0.5	0.2	0.1	0.0	0.0	0.2	0.2	0.2	(17)	
(18)		PrecStd	3.9	0.6	0.4	0.7	1.3	1.1	0.9	0.7	0.4	0.6	1.2	1.0	0.6	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	60.9	75.4	87.6	91.4	100.0	103.6	100.5	94.6	84.6	75.4	57.0	(19)	
(20)		0.4%	MCWB	40.7	46.5	53.2	61.7	62.0	63.3	65.3	64.5	61.4	57.9	54.1	44.4	(20)	
(21)		2%	DB	44.9	51.9	70.0	81.0	87.7	95.3	100.0	97.2	91.0	78.5	66.0	49.9	(21)	
(22)		2%	MCWB	37.5	42.7	51.1	58.3	60.9	63.4	66.1	63.7	60.1	54.9	50.3	41.1	(22)	
(23)		5%	DB	41.2	46.7	66.0	76.9	84.2	92.9	96.8	95.0	86.2	73.5	59.4	45.0	(23)	
(24)		5%	MCWB	35.5	39.8	50.4	57.1	60.3	63.1	64.4	62.7	59.1	53.5	47.0	38.2	(24)	
(25)		10%	DB	37.5	42.6	60.6	71.9	80.7	89.6	93.6	91.7	82.5	69.1	55.0	41.0	(25)	
(26)		10%	MCWB	33.3	37.6	48.2	54.7	59.4	62.7	63.5	62.4	58.0	51.7	44.7	35.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	48.3	56.2	65.0	67.1	70.7	73.1	71.1	65.3	58.9	55.3	45.4	(27)	
(28)		0.4%	MCDWB	49.4	59.6	70.4	81.1	83.3	89.2	96.1	91.8	87.0	74.9	71.3	53.7	(28)	
(29)		2%	WB	38.6	43.4	53.0	61.0	64.2	67.5	69.3	67.5	62.3	56.5	51.2	41.6	(29)	
(30)		2%	MCDWB	43.9	50.2	65.5	75.3	79.0	85.8	89.0	88.4	84.2	73.3	63.4	48.4	(30)	
(31)		5%	WB	35.8	40.2	50.9	58.9	62.3	65.5	66.9	65.5	60.6	54.9	48.2	38.7	(31)	
(32)		5%	MCDWB	40.0	45.7	62.6	70.7	77.5	84.0	88.8	86.7	81.3	69.2	58.2	44.4	(32)	
(33)		10%	WB	33.6	37.7	48.9	56.8	60.9	64.0	65.3	63.8	58.9	53.1	45.5	36.0	(33)	
(34)		10%	MCDWB	37.2	42.1	59.6	67.7	76.1	82.3	87.4	85.0	79.2	65.6	52.9	40.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.6	17.0	20.4	22.0	23.9	25.1	26.0	26.7	26.2	23.5	18.9	16.7	(35)	
(36)		5% DB	MCDBR	20.7	22.4	28.5	30.5	28.9	29.3	30.1	30.1	30.9	30.4	27.4	22.3	(36)	
(37)		5% DB	MCWBR	14.4	14.4	13.8	12.7	10.1	8.4	8.8	9.1	10.3	12.4	14.3	14.7	(37)	
(38)		5% WB	MCDWR	18.7	20.4	24.8	25.0	25.3	25.8	26.5	27.1	28.2	25.4	24.0	20.1	(38)	
(39)		5% WB	MCWBR	13.2	13.4	13.0	11.6	10.3	9.2	9.6	10.2	11.0	11.3	13.1	13.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.345	0.389	0.451	0.477	0.451	0.451	0.443	0.412	0.396	0.419	0.391	0.340	(40)		
(41)		taud	2.160	2.072	1.954	1.929	2.069	2.109	2.142	2.217	2.202	2.043	2.066	2.157	(41)		
(42)		Ebn at Noon	239	247	246	251	262	262	263	267	262	232	216	227	(42)		
(43)		Edh at Noon	31	41	52	57	51	49	47	43	40	41	34	29	(43)		
(44)	All-Sky Solar Radiation	RadAvg	644	937	1260	1599	1993	2169	2177	2022	1605	1008	636	519	(44)		
(45)		RadStd	53	79	103	148	143	112	123	102	77	111	67	42	(45)		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.48	N/A	N/A	N/A	N/A	

Nomenclature: See separate page